

# Set A

## Question - 1

(a)

$$I = \frac{V_{\max} - V_{\min}}{3R/2 + R + R + R/2} = \frac{8 - 0}{4R}$$

$$\therefore I = \frac{2}{R}$$

$$\therefore V_1 = R/2 \times I = R/2 \times \frac{2}{R} = 1V$$

$$V_2 = (R + R/2) \times I = 3R/2 \times \frac{2}{R} = 3V$$

$$V_3 = (R + R + R/2) \times I = \frac{5R}{2} \times \frac{2}{R} = 5V$$

| Quan. range | Quan. level | Encoded binary |
|-------------|-------------|----------------|
| 0-1         | 0.5         | 00             |
| 1-3         | 2           | 01             |
| 3-5         | 4           | 10             |
| 5-8         | 6.5         | 11             |

(b) Given, Max<sup>m</sup> digital output in the DAC  
→  $V_{out}$  at '11' = -6.5 volts

For R-2R ladder →

$$V_{out} = -\frac{V_{Ref} \times R_F}{2R} \left[ b_1 + \frac{b_0}{2} \right]$$

$$\Rightarrow -6.5 = \frac{-5 \times 13}{2R} \left[ 1 + \frac{1}{2} \right]$$

$$\Rightarrow R = \frac{-5 \times 13}{-6.5 \times 2} \times \frac{3}{2}$$

$$\therefore R = 7.5 \text{ k}\Omega$$

∴ Step size of the DAC →

$$V_{out} (\text{at } 01) = \frac{-5 \times 13}{2 \times 7.5} \left[ 0 + \frac{1}{2} \right]$$

$$\therefore \Delta_{R-2R} = -2.1667 \text{ volts}$$

∴ all output combinations →

| binary input (b <sub>1</sub> b <sub>0</sub> ) | analog output |
|-----------------------------------------------|---------------|
| 00                                            | 0 volts       |
| 01                                            | -2.1667 volts |
| 10                                            | -4.333 volts  |
| 11                                            | -6.5 volts    |

(c)

if  $V_{in} = 5.5$  volts given in the

ADC →

it falls under "5-8" Quantization range,

so, 5.5 volts → 11  
in binary

when '11' binary input is given to DAC

so, '11' binary input → -6.5 volts

(ans)

(d) clock frequency = 2 MHz

$$\therefore \text{Time period, } T_c = \frac{1}{2 \times 10^6} \text{ sec}$$

$$= 5 \times 10^{-7} \text{ s}$$

$$= 0.5 \mu\text{s}$$

$$\therefore T_1 = 2^N \times T_c$$

$$= 2^2 \times 0.5 \mu\text{s}$$

$$= 2 \mu\text{s}$$

$$\therefore T_2 = 2.7 \mu\text{s} - 2 \mu\text{s}$$

$$= 0.7 \mu\text{s}$$

Now,

$$T_2 = T_1 \cdot \frac{V_{in}}{V_{Ref}}$$



$$\Rightarrow V_{in} = \frac{T_2 \cdot V_{ref}}{T_1}$$

$$= \frac{0.7 \mu s \times 8 V}{2 \mu s}$$

$$V_{in} = 2.8 \text{ volts}$$

$$\text{Max}^m \text{ conversion time} = 2 \times T_1$$

$$[\text{when, } T_1 = T_2] \leftarrow = 4 \mu s$$

(e) Input voltage range = 8V

$$\text{number of steps} = 2^N = 2^2$$

$$= 4$$

$$\text{step size} = \frac{8}{4} = 2 \text{ volts}$$

$$V_{in} = m \times \text{step size}$$

$$\Rightarrow m = \frac{V_{in}}{\text{step-size}} = \frac{5.5}{2} = 2.75 \approx 2$$

| 'm' value | Q. range | binary |
|-----------|----------|--------|
| $m = 0$   | 0 - 2    | 00     |
| $m = 1$   | 2 - 4    | 01     |
| $m = 2$   | 4 - 6    | 10     |
| $m = 3$   | 6 - 8    | 11     |

Thus, output of ADC will be '10'  
which will be given to R2R DAC →

$$Y_{out}(\text{at } 10) = \frac{-5 \times 13}{2 \times 7.5} \left[ 1 + \frac{0}{2} \right]$$

$$= -4.33 \text{ volts}$$

$$\% \text{ error} = \left| \frac{-6.5 - (-4.33)}{-6.5} \right| \times 100\%$$

$$\% \text{ error} = 33.38\%$$

## Question-2

(a) The triangular wave generator has non inverting Schmitt trigger

$$V_{TH} = -\frac{R_1}{R_2} V_L \quad ;$$

$$= -\frac{2}{8} \times (-8)$$

$$= 2 \text{ V}$$

$$V_{TL} = -\frac{R_1}{R_2} V_H$$

$$= -\frac{2}{8} \times (4)$$

$$= -1 \text{ V}$$

$$T_1 = R_i \cdot C_i \left[ \frac{V_{TH} - V_{TL}}{V_H} \right]$$

$$= R_i \cdot C_i \left[ \frac{2 - (-1)}{4} \right] = \frac{3}{4} R_i \cdot C_i$$

$$T_2 = R_i \cdot C_i \left[ \frac{V_{TL} - V_{TH}}{V_L} \right]$$

$$= R_i \cdot C_i \left[ \frac{-1 - 2}{-8} \right] = \frac{3}{8} R_i \cdot C_i$$

$$\therefore \text{Time period, } T = T_1 + T_2 = \left( \frac{3}{4} + \frac{3}{8} \right) R_i \cdot C_i$$

$$= \frac{9}{8} R_i \cdot C_i$$

$$\therefore \text{frequency, } f = \frac{1}{T} = \frac{8}{9 R_i \cdot C_i}$$

(Proved)

(b) To figure out the duty cycle  $\rightarrow$

The triangular wave generator's duty

at  $V_B \rightarrow$



is a square  
wave

$$D.C = \frac{T_1}{T_1 + T_2} \times 100\%$$

$$= \frac{\frac{3}{4} R_1 \cdot C_1}{\frac{9}{8} R_1 \cdot C_1} \times 100\%$$

$$= 0.67 \times 100\%$$

$$= 67\%$$

This duty cycle will be applied to the  
Square wave generator, which is to  
be designed  $\rightarrow$

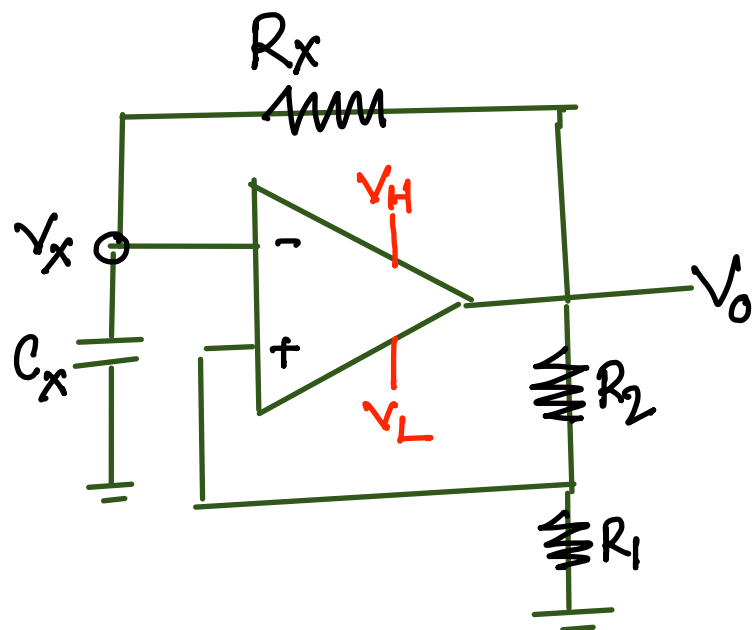
$$R_1 = 10k\Omega$$

$$R_2 = 20k\Omega$$

$$R_x = 10k\Omega$$

$$C_x = 50nF$$

$$f = 1kHz$$



$$\therefore \text{Time period, } T = \frac{1}{f} = \frac{1}{1 \times 10^3} \text{ sec} \\ = 1 \text{ ms}$$

$$\therefore \text{High duration, } T_1 = (\text{D.C.}) \times T \\ = 0.67 \times 1 \text{ ms} \\ = 0.67 \text{ ms}$$

$$\therefore \text{Low duration, } T_2 = T - T_1 \\ = 0.33 \text{ ms}$$

$$\therefore \text{Time constant, } T = R_x \cdot C_x \\ = (10 \text{ k}\Omega) \times (50 \text{ nF}) \\ = 5 \times 10^{-4} \text{ sec} \\ = 0.5 \text{ ms}$$

$$\text{Now, } T_1 = T \ln \frac{V_H - V_{TL}}{V_H - V_{TH}}$$

$$\frac{R_1}{R_1 + R_L} = \frac{10}{10 + 20} \\ = \frac{1}{3}$$

$$\Rightarrow \frac{T_1}{J} = \ln \frac{V_H - \left(\frac{R_1}{R_1+R_2}\right) \cdot V_L}{V_H - \left(\frac{R_1}{R_1+R_2}\right) V_H}$$

$$\Rightarrow e^{\frac{0.67}{0.5}} = \frac{V_H - \frac{1}{3}V_L}{V_H - \frac{1}{3}V_H}$$

$$\Rightarrow 3.82 \times \left[ \frac{2}{3}V_H \right] = V_H - \frac{1}{3}V_L$$

$$\Rightarrow 2.546V_H - V_H = -0.33V_L$$

$$\Rightarrow 1.546V_H = -0.33V_L$$

$$\therefore V_L = -4.685V_H$$

let's assume, positive bias voltage

$$\rightarrow V_H = 10 \text{ V}$$

the negative bias

$$\text{voltage, } V_L = -4.685 \times 10$$

$$= -46.85 \text{ volts}$$

# Question - 3 $\Rightarrow$

(a) From figure-4, we can see that it's a non-inverting schmitt trigger output,

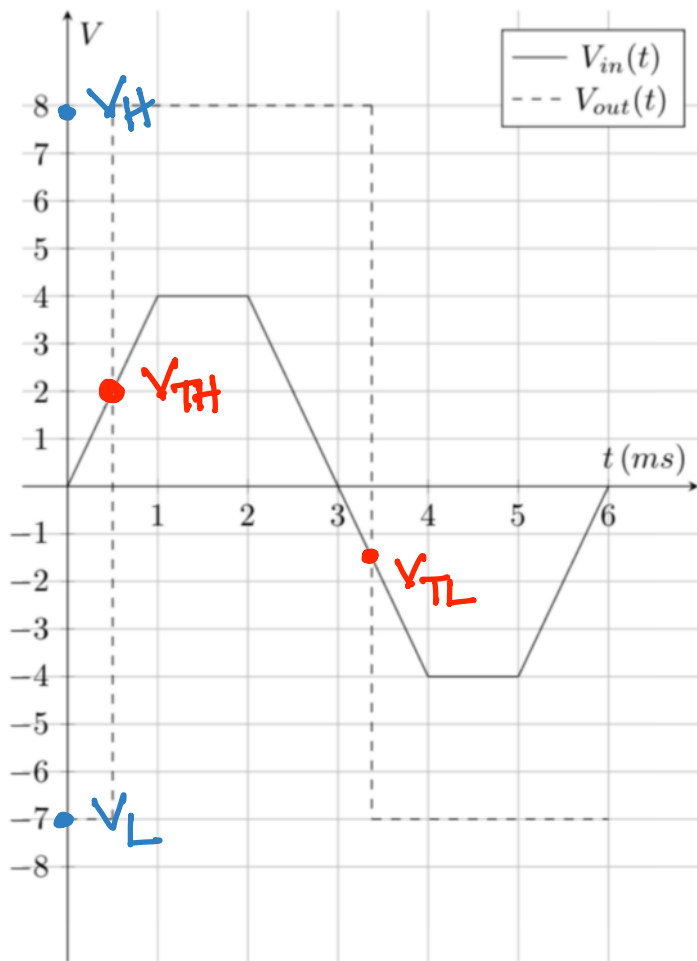


Figure 4:  $V_{in}(t)$  &  $V_{out}(t)$

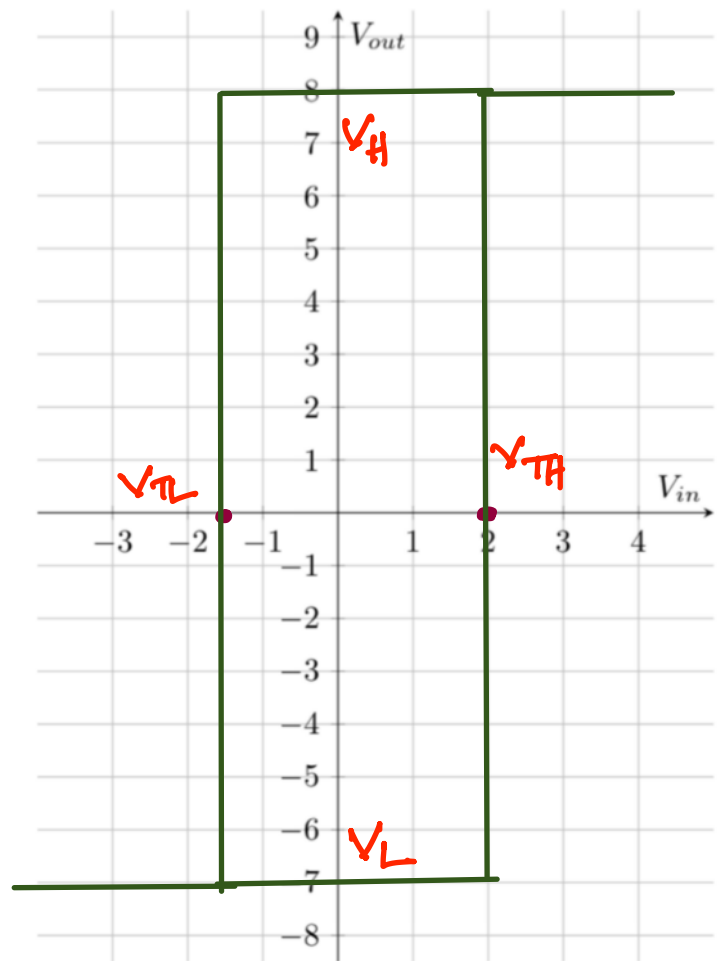


Figure 5: VTC of the Schmitt Trigger

$\therefore$  bias voltages  $\begin{cases} V_H = 8V \\ V_L = -7V \end{cases}$

$\therefore$  Threshold voltages  $\begin{cases} V_{TH} = 2V \\ V_{TL} = -1.5V \end{cases}$



(b) We know, for non-inv. Schmitt trigger  $\rightarrow$

$$V_{TH} = -\frac{R_1}{R_2} V_L + \frac{R_1 + R_2}{R_2} V_{Ref}$$

shift voltage  
' $V_s$ '

$$V_{TL} = -\frac{R_1}{R_2} V_H + \frac{R_1 + R_2}{R_2} V_{Ref}$$

$$V_{TH} - V_{TL} = \frac{R_1}{R_2} (V_H - V_L) + 0$$

$$\Rightarrow \frac{V_{TH} - V_{TL}}{V_H - V_L} = \frac{R_1}{R_2}$$

$$\Rightarrow \frac{2 - (-1.5)}{8 - (-7)} = \frac{R_1}{R_2}$$

$$\Rightarrow \frac{3.5}{15} = \frac{R_1}{R_2}$$

$$\therefore R_2 = 4.285 R_1$$

let's assume

$$R_1 = 1k\Omega$$

$$\therefore R_2 = 4.285k\Omega$$

Now,

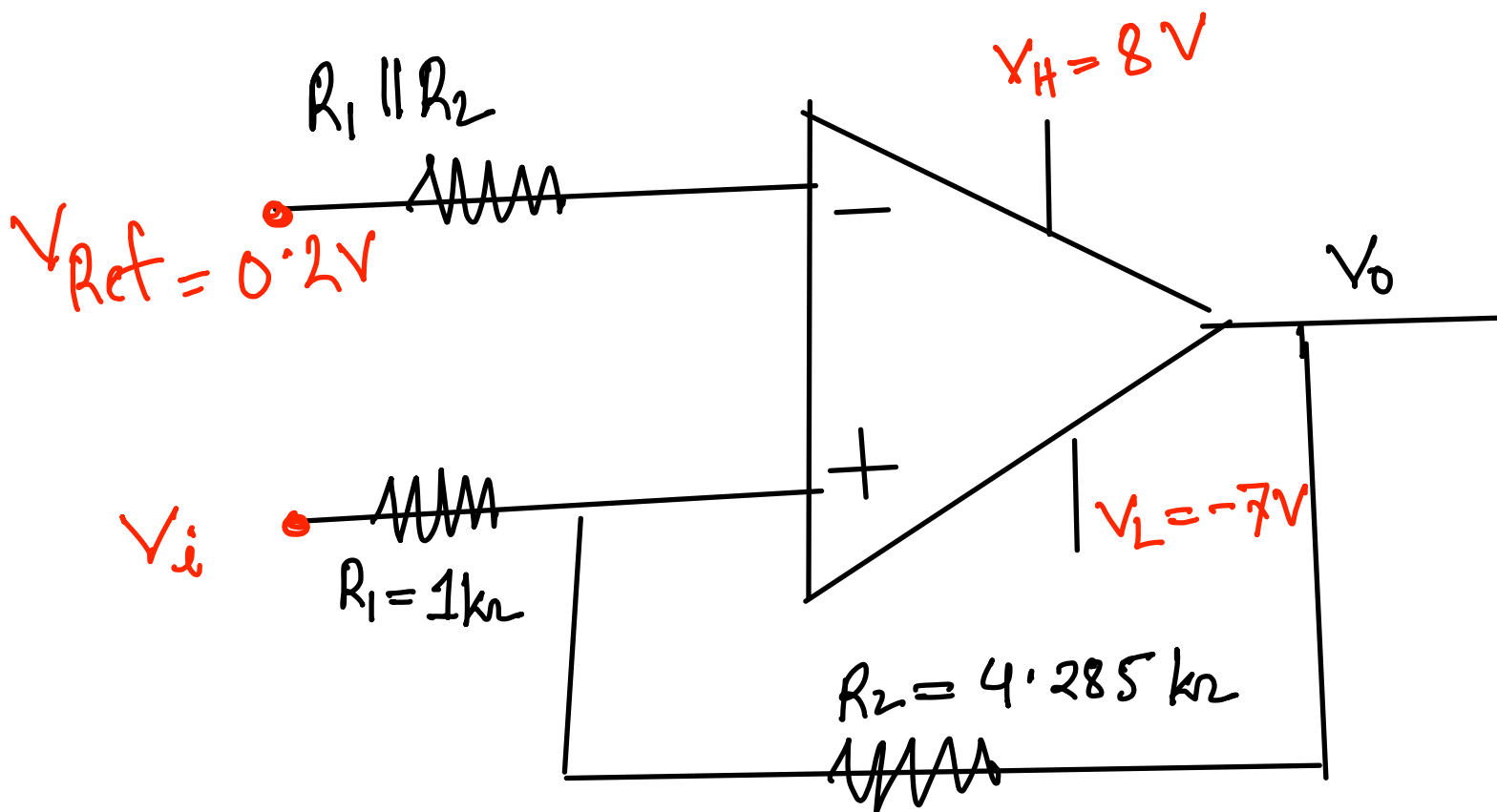
$$V_S = \frac{R_1 + R_2}{R_2} V_{Ref}$$

$$\Rightarrow \frac{V_{TH} + V_{TL}}{2} = \frac{R_1 + R_2}{R_2} V_{Ref}$$

$$\Rightarrow \frac{2 - 1.5}{2} = \frac{R_1 + 4.285 R_1}{4.285 R_1} \cdot V_{Ref}$$

$$\therefore V_{Ref} = 0.2 \text{ volts}$$

The design  $\rightarrow$



(C) Interchanging ' $V_i$ ' and ' $V_{ref}$ ' will give us  $\rightarrow$  inverting Schmitt trigger

Thus,

$$V_{TH} = \frac{R_1}{R_1 + R_2} V_H + \frac{R_2}{R_1 + R_2} V_{ref}$$

$$= \frac{1}{1 + 4.285} \times 8 + \frac{4.285}{1 + 4.285} \times 0.2$$

$$\therefore V_{TH} = 1.675 \text{ volts}$$

$$V_{TL} = \frac{R_1}{R_1 + R_2} V_L + \frac{R_2}{R_1 + R_2} V_{ref}$$

$$= \frac{1}{1 + 4.285} \times (-7) + \frac{4.285}{1 + 4.285} \times 0.2$$

$$\therefore V_{TL} = -1.162 \text{ volts}$$

$$\therefore \text{Hysteresis width, } V_{HW} = V_{TH} - V_{TL} = 2.837 \text{ V}$$

$$\therefore \text{Shift voltage, } V_s = \frac{V_{TH} + V_{TL}}{2} = 0.2565 \text{ volts}$$

## Question - 4

(a) Practically, Not possible

Theoretically,

$$V_{HW} = V_{TH} - V_{TL} \Rightarrow 0 = V_{TH} - V_{TL}$$

$$\therefore V_{TH} = V_{TL}$$

for that, it needs to be

$$V_H = V_L$$

(b) as,  $t_1 = T_c \times 2^N$  (all terms are constant)

$t_1$  doesn't depend on  $V_{in}$

→ that's why remains constant

and,  $t_2 = t_1 \cdot \frac{V_{in}}{V_{Ref}}$  → constant

$t_2 \propto V_{in}$  → that's why changes

(c)

Wave 'A' →

non-inverting terminal  
of the Schmitt trigger  
in triangular wave gene.

Wave 'B' →

inverting terminal  
in square wave gene.